

Paper Code: ITC10

Subject: **Probability and Stochastic Processes**

Time: 1.5 Hours

Note: Attempt all questions.

Maximum Marks: 25

Q. No.	Questions	CO	Marks
1.	Let X be a uniform random variable over $(-1,1)$. Let $Y = X^n$. (a) Calculate the covariance of X and Y . (b) Calculate the correlation coefficient of X and Y .	CO1	5
2.	Show that if $Y = X^2$, then $f_Y(y X \geq 0) = \frac{U(y)}{1 - F_X(0)} \frac{f_X(\sqrt{y})}{2\sqrt{y}}$	CO3	5
3.	A noisy resistor produces a voltage $v_n(t)$. At $t=t_1$, the noise level $X \triangleq v_n(t_1)$ is known to be Gaussian random variable with pdf $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left[-\frac{1}{2}\left(\frac{x}{\sigma}\right)^2\right]$ Compute the probability that $ X > k\sigma$ for $k=0, 1, 2, \dots$	CO2	5
4.	Random variables X and Y are independent with $f_X(x) = \frac{x}{\alpha^2} e^{-x^2/2\alpha^2} U(x) \text{ and } f_Y(y) = \begin{cases} \frac{1}{\pi\sqrt{1-y^2}} & , y < 1 \\ 0 & , y > 1 \end{cases}$ Show that $Z=XY$ is Gaussian distributed with 0 mean and variance α^2 .	CO2	5
5.	Assume that the length of a phone call in minutes is an exponential random variable X with parameter $\lambda = \frac{1}{10}$. If someone arrives at a phone booth just before you arrive, find the probability that you have to wait (a) less than 5 minutes and (b) between 5 and 10 minutes.	CO3	5

THIRD SEMESTER B. Tech. Course work
END SEMESTER EXAMINATION, December - 2021
Course Code- EAEPC06
Course Title- Probability Theory and Random Process
(For East Campus, ECAM students)

Roll no.

Time- 3 Hours

Max. Marks- 50

Note: - Attempt all the five questions. Missing data/ information if any, maybe suitably assumed & mentioned in the answer.

Q. No	Question	Marks	CO
Q 1	Attempt any 2 parts of the following.		
1a	Describe the cumulative distribution function (CDF) and probability density function (pdf) of a random variable. What is the relation between the CDF and the pdf?	5	CO2
1b	Describe Gaussian distribution. Derive its mean and variance.	5	CO4
1c	Describe uniform distribution. Derive its mean and variance.	5	CO4
Q 2	Attempt any 2 parts of the following.		
2a	Describe joint distribution function and joint density function.	5	CO2
2b	Define ij -th central moment of two random variables, X and Y . Write down the expression of covariance of X and Y and explain its significance.	5	CO2
2c	Explain conditional distribution and conditional expectation.	5	CO3
Q 3	Attempt any 2 parts of the following.		
3a	Consider X , a random variable with the pdf $f_X(x)$. Derive the pdf of Y where $Y = g(X)$.	5	CO4
3b	The pdf of a random variable X is given by $f_X(x) = 1$, for $0 < x \leq 1$, and zero elsewhere. Determine the pdf of Y where $Y = 3X + 2$.	5	CO4
3c	State and prove Chebyshev inequality.	5	CO4
Q 4	Attempt any 2 parts of the following.		
4a	Describe Wide Sense Stationary (WSS) process.	5	CO3
4b	Describe autocorrelation function of random process and its properties.	5	CO3
4c	The autocorrelation function of a random process is given by, $R_{xx}(\tau) = 25 + \frac{4}{1+6\tau^2}$. Find mean value and variance of the process.	5	CO4
Q 5	Attempt any 2 parts of the following.		
5a	Derive the relationship between the power density spectrum and the autocorrelation function of a signal.	5	CO4
5b	Derive the power density spectrum of the output of the linear time invariant system having a transfer function $H(\omega)$. The power density spectrum of the input process is $S_{xx}(\omega)$.	5	CO4
5c	Describe white noise along with its power density spectrum and the autocorrelation function.	5	CO3

Total no. of Pages:

Roll. No:.....

NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY
THIRD SEMESTER-B.TECH.(ITIOT)

END-SEMESTER EXAMINATION, December, 2021(West Campus)

Course Code: IIEPC06

Course Title: Probability and stochastic process

Time: 3 Hours

Max. Marks: 50

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

S.No.	Question	Marks	CO
1.	Attempt any two parts		
(a)	Calculate mean, mean square value and variance of uniform distribution over $(0, \pi)$	5	CO1
(b)	Let $Y = aX + b$. Show that if $X = N(\mu, \sigma^2)$, Then prove that $Y = N(a\mu + b, a^2 \sigma^2)$.	5	
(c)	Binary data are transmitted over a noisy communication channel in block of 16 binary digits. The probability that a received digit is in error due to channel noise is 0.01. Assume that the errors occurring in various digit positions within a block are independent. (i) Find the mean errors per block. (ii) Find the variance of the number of error per block. (iii) Find the probability that the number of errors per block is greater than or equal to 4.	5	
2.	Attempt any two parts		
(a)	Let X and Y be Joint continuous random variables with joint PDF is given below $f_{XY}(x, y) = \begin{cases} 2 & x, y > 0 \text{ and } x + y \leq 1 \\ 0 & \text{otherwise} \end{cases}$ Find co-variance of X and Y	5	CO2
(b)	Find Moment generating function of a random variable X uniformly distributed over (a, b) also calculate $E[X]$.	5	
(c)	Let $Z = aX + bY$, where a and b are arbitrary constants. Show that if X and Y are independent, then $\sigma_z^2 = a^2 \sigma_x^2 + b^2 \sigma_y^2$	5	

3	Attempt any two parts		
(a)	State and prove the 'Chebyshev inequality'.	5	CO3
(b)	If X is Uniformly distributed in $[-\pi/2, \pi/2]$. Find the pdf of $Y = \tan X$.	5	
(c)	Show that if X and Y are zero mean jointly normal random variables, then $E[X^2 Y^2] = E[X^2]E[Y^2] + 2(E[XY])^2$	5	
4.	Attempt any two parts		
(a)	The autocorrelation function of a random process is given below. Calculate mean and variance of random process. $R_{XX}(\tau) = 25 + \frac{4}{1+6\tau^2}$	5	CO4
(b)	Consider a random process $X(t)$ defined by: $X(t) = \cos(At)$ Where A is a random Variable uniformly distributed over $[0, W_0]$. Determine whether $X(t)$ is stationary or not.	5	
(c)	Consider the random process $X(t)$ defined by: $X(t) = A \cos(wt)$ Where 'w' is a constant and 'A' is a random variable uniformly distributed over $[0, 1]$. Find the autocorrelation and autocovariance of $X(t)$.	5	
5.	Attempt any two parts		
(a)	Describe the Markovian queuing model (M/M/1, M/M/1/M) in brief.	5	CO5
(b)	Let X_n be an iid-valued random process. Show that X_n is a Markov process and give its one-step transition probability matrix.	5	
(c)	Discuss the probability distribution function (PDF) and Cumulative distribution function (CDF), Also discuss the properties of CDF and PDF.		CO1

III SEMESTER B.Tech (IT, ITNS)
END-SEMESTER EXAMINATION 2021

Course Code- ITECC06/ INECC06 Course Title- Probability Theory and Stochastic Process.

Time: 3:00 Hrs

Max Marks: 50

Attempt all questions. Missing data / information (If any) may be suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO												
Q1	Attempt any 2 out of 3 parts In a box there are 100 resistors having resistance and tolerance as given in table. Let a resistor be selected from the box and assume each resistor has the same likelihood of being chosen. Define three events: A as draw a 47 ohm resistor, B as draw a resistor with 5% tolerance and C as draw a 100 ohm resistor. a <table border="1"> <thead> <tr> <th>Resistance</th> <th>Tolerance 5%</th> <th>Tolerance 10%</th> </tr> </thead> <tbody> <tr> <td>22</td> <td>10</td> <td>14</td> </tr> <tr> <td>47</td> <td>28</td> <td>16</td> </tr> <tr> <td>100</td> <td>24</td> <td>8</td> </tr> </tbody> </table> Find the conditional probabilities of $P(A B)$, $P(A C)$ and $P(B C)$. b A submarine attempts to sink an aircraft carrier. It will be successful only if two or more torpedoes hit the carrier. If the submarine fires three torpedoes and the probability of a hit is 0.4 for each torpedo. What is the probability that the carrier will be sunk. c An elementary binary communication system consists of a transmitter that sends one of two possible symbols (a 1 or a 0) over a channel to a receiver. The probabilities that the symbol 1 and 0 are selected for transmission are assumed to be $P(1) = 0.6$ and $P(0) = 0.4$. The channel occasionally causes errors to occur so that a '1' shows up at the receiver as a '0', and vice versa. The channel is diagrammatically shown as Find the probabilities of system errors.	Resistance	Tolerance 5%	Tolerance 10%	22	10	14	47	28	16	100	24	8	05	CC1
Resistance	Tolerance 5%	Tolerance 10%													
22	10	14													
47	28	16													
100	24	8													
		05	CO1												
		05	CO1												
Q2	Attempt any 2 out of 3 parts														

a	Let X be a random variable with cdf $F_X(x) = \begin{cases} 1, & x \geq 1 \\ \frac{1}{2} + \frac{x}{2}, & 0 \leq x < 1 \\ 0, & x < 0 \end{cases}$ a. Find the pdf of X. b. Find the $E(e^x)$. c. Find $P(X=0 X \leq 0.5)$	05	CO2
b	A binary source generates digits <u>1</u> and <u>0</u> randomly with probabilities <u>0.6</u> and <u>0.4</u>, respectively. a) What is the probability that two 1's and three 0's will occur in a five-digit sequence? b) What is the probability that at least three 1's will occur in a five-digit sequence?	05	CO2
c	Let X be a continuous random variable with pdf given by $f_X(x) = \frac{1}{2} e^{- x } \text{ for all } x \in \mathbb{R}$ If $Y = X^2$, find the cdf of Y.	05	CO2
Q3 Attempt any 2 out of 3 parts			
a	Consider an experiment of drawing randomly three balls from an urn containing two red, three white, and four blue balls. Let (X, Y) be a bivariate r.v. where X and Y denote, respectively, the number of red and white balls chosen. a) Find the range of (X, Y). b) Find the joint pmf's of (X, Y). c) Find the marginal pmf's of X and Y. d) Are X and Y Independent.	05	CO3
b	Let X and Y be r.v.'s with joint pdf $f_{XY}(x, y)$ and joint cdf $F_{XY}(x, y)$. Let $W = \min(X, Y)$ where X and Y are exponential distributions with parameters (λ, μ). a) Find the cdf of W. b) Find the pdf of W if X and Y are independent.	05	CO3
c	Let (X, Y) be a bivariate r.v. with the joint pdf $f_{XY}(x, y) = \frac{x^2 + y^2}{4\pi} e^{-(x^2 + y^2)/2} \quad -\infty < x < \infty, -\infty < y < \infty$ Comment whether X and Y are uncorrelated or not. Give proper justification.	05	CO3
Q4 Attempt any 2 out of 3 parts			

	c) Find the marginal pmf's of X and Y. d) Are X and Y Independent.		
b	Let X and Y be r.v.'s with joint pdf $f_{XY}(x, y)$ and joint cdf $F_{XY}(x, y)$. Let $W = \min(X, Y)$ where X and Y are exponential distributions with parameters (λ, μ) . a) Find the cdf of W. b) Find the pdf of W if X and Y are independent.	05	CC3
c	Let (X, Y) be a bivariate r.v. with the joint pdf $f_{XY}(x, y) = \frac{x^2 + y^2}{4\pi} e^{-(x^2 + y^2)/2} \quad -\infty < x < \infty, -\infty < y < \infty$. Comment whether X and Y are uncorrelated or not. Give proper justification.	05	CO3
Q4	Attempt any 2 out of 3 parts		
a	Let two random processes X(t) and Y(t) be defined by $X(t) = A \cos(w_0 t) + B \sin(w_0 t)$ $Y(t) = B \cos(w_0 t) - A \sin(w_0 t)$ where, A and B are uncorrelated zero mean random variable with same variance σ^2 and w_0 is a constant. Consider X(t) and Y(t) are WSS process. Find the cross-correlation $R_{XY}(t, t + \tau)$ function.	05	CC4
b	Let a random process X(t) be defined by $X(t) = A \cos(w_0 t + \theta)$ where, A and w_0 are constant and θ is uniformly distributed random variable over $(-\pi, \pi)$. Show that X(t) is WSS.	05	CO4
c	Let X_n be a sequence of independent Bernoulli random variables. X_n is then an iid random process taking on values from the set {0, 1}. An UP-Down counter is driven by +1 or -1. Let the input to the counter be given by $D_n = 2X_n - 1$. Then $D_n = \begin{cases} +1 & \text{if } X_n = 1 \\ -1 & \text{if } X_n = 0 \end{cases}$ Find the mean and variance of D_n .	05	CO4
Q5	Attempt any 2 out of 3 parts		

c	Let X_n be a sequence of independent Bernoulli random variables. X_n is then an iid random process taking on values from the set $\{0, 1\}$. An UP-Down counter is driven by $+1$ or -1. Let the input to the counter be given by $D_n = 2X_n - 1$. Then $D_n = \begin{cases} +1 & \text{if } X_n = 1 \\ -1 & \text{if } X_n = 0 \end{cases}$ Find the mean and variance of D_n.	05	C
Q5	Attempt any 2 out of 3 parts		
a	In the M/M/1 queuing system, the arrival process is the Poisson process with rate λ (mean arrival rate) and the service time is exponentially distributed with parameter μ (mean service time). Derive the expression for (a) Average number of customers in the system and (b) Average number of customers waiting in the queuing.	05	C
b	Customers arrive at watch repair shop according to a Poisson process at a rate of 1 per every 10 minutes, and the service time is an exponential random variable with mean 8 minutes. (a) Find the average time (W) a customer spends in the shop, and the average time a customer spends in waiting for service (W_q). (b) Suppose that the arrival rate of the customers increases by 10%. Find the corresponding changes in W and W_q.	05	C
c	Let $X(t)$ be continuous time Gaussian random process with mean and covariance given by following functions $m_X = 3t, \quad C_X(t_1, t_2) = 9e^{2 t_1 - t_2 }$ Find $P[X(3) < 6]$ and $P[X(1) + X(2) > 2]$.	05	C

Date:

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02/2022

III SEMESTER B.Tech (ECE, EIOT)

END-SEMESTER EXAMINATION 2021

Course Code- ECECC06/ EIECC06 Course Title- Probability Theory and Random Processes

Time: 3:00 Hrs

Max Marks: 50

Attempt all questions. Missing data / information (if any) may be suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO
Q1	Attempt any 2 out of 3 parts		
a	The lifetime of the system expressed in weeks is Rayleigh random variable X for which $f_X(x) = \begin{cases} (x/200)e^{-x^2/400} & 0 \leq x \\ 0 & x < 0 \end{cases}$ Determine the expected lifetime of the system given that the system has survived <u>20 weeks</u>.	05	CO1
b	A continuous random variable X has the PDF $f_X(x) = \begin{cases} \frac{1}{3}e^{-x/3}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$ (a) Find moment generating function of X. (b) Find E[X] and var[X].	05	CO1
c	Let X be the random variable with the following pdf. $f_X(X) = \frac{1}{2}e^{- x } \quad \text{for all } x$ Find CDF of Y if Y=X²	05	CO1
Q2	Attempt any 2 out of 3 parts		
a	The radial "miss distance" of landing from parachuting sky divers as measured from the target center is random variable with distribution function $F_X(x) = [1 - e^{-x^2/800}]u(x)$ The target is a circle of 50-meter radius with bull's eye of 10meter radius. Find the probability of hitting the bull's eye given that the landing is on the target.	05	CO2
b	Random variables X1 and X2 have the joint characteristic function $\phi_{X_1, X_2}(\omega_1, \omega_2) = [(1 - j2\omega_1)(1 - j2\omega_2)]^{-N/2}$ where N>0 is an integer.	05	CO2

$$m_{10} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{x^2} dx$$

$$m_{10} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{x^2} dx$$

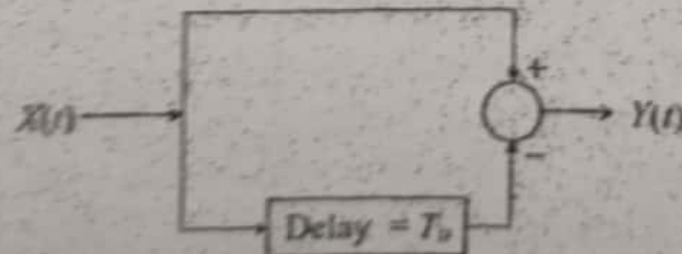
Determine the following				
(i) m_{20} and				
(ii) Correlation coefficient:				
Let X and Y be Joint continuous random variables with joint PDF is given below				
c	$f_{XY}(x,y) = \begin{cases} 2 & x, y > 0 \text{ and } x+y \leq 1 \\ 0 & \text{otherwise} \end{cases}$		05	CO2
(a) Find co-variance of X and Y				
(b) Find Co-relation coefficient of X and Y				
Q3 Attempt any 2 out of 3 parts				
a	A Gaussian random variable for which $\bar{X}=0.6$ and $\sigma_X=0.8$ is transformed to a new random variable by transformation $Y = T(X) = \begin{cases} 4 & 1.0 \leq X < \infty \\ 2 & 0 \leq X < 1.0 \\ -2 & -1.0 \leq X < 0 \\ -4 & -\infty < X < -1.0 \end{cases}$		05	CO3
Find the density function of Y.				
b	If X is Uniformly distributed in $[-\pi/2, \pi/2]$. Find the p.d.f of random variable Y if the transformation is $Y=\tan X$.		05	CO3
c	If X and Y are independent random variables having PDFs $f_X(x) = 2e^{-2x}u(x) \text{ and } f_Y(y) = 3e^{-3y}u(y).$		05	CO3
Find the density function of their sum $Z = X + Y$.				
Q4 Attempt any 2 out of 3 parts				
a	Consider a random process $X(t)$ given by $A\cos(2\pi f_c t + \theta)$ where f_c and θ are constant where A is a Gaussian random variable with zero mean and variance σ^2. Determine whether		05	CO3

c	(a) Find co-variance of X and Y. (b) Find Co-relation coefficient of X and Y		
Attempt any 2 out of 3 parts			
Q3	A Gaussian random variable for which $\bar{X}=0.6$ and $\sigma_X=0.8$ is transformed to a new random variable by transformation	05	CO3
a	$Y=T(X)=\begin{cases} 4 & 1.0 \leq X < \infty \\ 2 & 0 \leq X < 1.0 \\ -2 & -1.0 \leq X < 0 \\ -4 & -\infty < X < -1.0 \end{cases}$ Find the density function of Y.		
b	If X is Uniformly distributed in $[-\pi/2, \pi/2]$. Find the p.d.f of random variable Y if the transformation is $Y=\tan X$.	05	CO3
c	If X and Y are independent random variables having PDFs $f_X(x) = 2e^{-2x}u(x)$ and $f_Y(y) = 3e^{-3y}u(y)$. Find the density function of their sum $Z = X + Y$.	05	CO3
Attempt any 2 out of 3 parts			
Q4	a Consider a random process $X(t)$ given by $A\cos(2\pi f_c t + \theta)$ where f_c and θ are constant where A is a Gaussian random variable with zero mean and variance σ_A^2 . Determine whether $X(t)$ is wide sense stationary.	05	CO4
b	A random process is described by $X(t) = A\sin(t) + B\cos(t)$ where A and B are independent random variables with zero means and equal standard deviation. Show that the process is stationary of second order.	05	CO4
c	The impulse response of an LTI system is $ae^{-\alpha t}u(t)$ where $\alpha = \frac{1}{RC}$. If a zero mean white Gaussian process $N(t)$ is input into this filter, find the ACF of the output.	05	CO4
Attempt any 2 out of 3 parts			
Q5			

a	In a communication system, the carrier signal at the receiver is modeled by $X_t = \cos(2\pi f t + \phi)$, where $\phi \sim \text{uniform}[-\pi, \pi]$. Find the spectral density	05	CO5
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a	and θ are constant where A is a Gaussian random variable with mean and variance σ_A^2 . Determine whether $X(t)$ is wide sense stationary.		
b	A random process is described by $X(t) = A \sin(t) + B \cos(t)$ where A and B are independent random variables with zero means and equal standard deviation. Show that the process is stationary of second order.	05	CO4
c	The impulse response of an LTI system is $ae^{-\alpha t}u(t)$ where $\alpha = \frac{1}{RC}$. If a zero mean white Gaussian process $N(t)$ is input into this filter, find the ACF of the output.	05	CO4
Q5	Attempt any 2 out of 3 parts		

a	In a communication system, the carrier signal at the receiver is modeled by $X_c = \cos(2\pi f_c t + \theta)$, where $\theta \sim \text{uniform}[-\pi, \pi]$. Find the power spectral density	05	CO5
b	A white noise process $X(t)$ with zero mean and power spectral density $\frac{N_0}{2}$ is applied to a linear time invariant filter with impulse response $h(t)$, producing an output $Y(t)$. Find the cross-correlation function between X and Y .	05	CO5
c	A wide sense stationary random process $X(t)$ passes through the LTI system shown in the figure. If the autocorrelation function of $X(t)$ is $R_{XX}(\tau)$. Find the autocorrelation function of the output process $Y(t)$.	05	CO5



III SEMESTER B.Tech (IT, ITNS)
MID-SEMESTER EXAMINATION 2023

Course Code : INECC06/ITECC06	Time: 1:30 Hrs
Course Title : Probability & Stochastic Processes	Max Marks: 25
Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.	

Q. No.	Questions	Marks	CO
Q1	Given that $P(A) = 0.9$, $P(B) = 0.8$, and $P(A \cap B) = 0.75$, find (i) $P(A \cup B)$ (ii) $P(A \cap B^c)$ (iii) $P(A \cap B^c)$ and (iv) $P(A^c \cup B^c)$.	04	CO1
Q2	Let X be a random variable which assumes values x_1, x_2, x_3, x_4 such that $2P(X = x_1) = 3P(X = x_2) = P(X = x_3) = 5P(X = x_4)$. Find the CDF and PMF of X .	01	CO2
Q2	For any three events A_1, A_2 , and A_3 , show that $P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3) + P(A_1 \cap A_2 \cap A_3)$	04	CO1
Q2	Describe the Bernoulli and Poisson random variables.	01	CO2
Q3	The median of a continuous random variable X is the value of $x = x_0$ such that $P[X \geq x_0] = P[X \leq x_0]$. The mode of X is the value of $x = x_m$ at which the PDF achieves the maximum value. Find the median and Mode of i. Exponential random Variable. ii. Gaussian random Variable.	04	CO2

3	b	Find the constant 'b>0' so that the function $f_X(a) = \begin{cases} \exp\left(\frac{3a}{4}\right), & 0 \leq a \leq b \\ 0 & \text{elsewhere} \end{cases}$ Is a valid PDF.	01	CO2
Q 4	a	A random Variable X is gaussian distributed with $m_X = 0, \sigma_X^2 =$ 1. What is the probability that i. $ X > 2$ ii. $ X \geq 2$ iii. $X^2 > 9$ iv. $ 2X + 3 > 2$ v.	04	CO2
	b	A number is selected at random from (1,2,3100). Given that the selected number is divisible by 2, find the probability that it is divisible by 3 or 5?	01	CO1
Q 5	a	The joint PDF of random Variables X,Y is given by $f_{XY}(\alpha, \beta) = \begin{cases} k \exp(-[u\alpha + g\beta]), & \alpha > 0, \beta > 0 \\ 0 & \text{otherwise} \end{cases}$, where 'k', 'u>0' and 'g>0' are constants. i. Determine the value of 'k' ii. Are X, Y independent iii. Compute $P[X > Y]$, for $u = 10^{-4}, g = 1.2 \times 10^{-4}$ iv. Compute $P[X > 1]$, for $u = 10^{-4}, g = 1.2 \times 10^{-4}$ v.	04	CO3
	b	Define the Joint Gaussian probability Density Function.	01	CO3

Total no. of Pages:2

Roll. No:.....

NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY**THIRD SEMESTER-B.TECH.(ECE/EIOT)****MID-SEMESTER EXAMINATION, September, 2023**

Course Code: ECECC06/EIECC06 Course Title: Probability Theory and Random Processes

Time: 1.5 Hours

Max. Marks: 25

Note: Attempt all questions. Missing data/information(if any), may be suitably assumed & mentioned in the answer.

S.No.	Question	Marks	CO
1.	Attempt all parts		
(a)	Calculate mean, mean square value and variance of uniform distribution over $(0, \pi)$.	2	CO1
(b)	Binary data are transmitted over a noisy communication channel in block of 16 binary digits. The probability that a received digit is in error due to channel noise is 0.01. assume that the errors occurring in various digit positions within a block are independent. (i) Find the mean errors per block. (ii) Find the variance of the number of errors per block. (iii) Find the probability that the number of errors per block is greater than or equal to 4.	3	CO1
2.	Attempt all parts		
(a)	Let X and Y be Joint continuous random variables with joint PDF is given below $f_{XY}(x, y) = \begin{cases} 2 & x, y > 0 \text{ and } x + y \leq 1 \\ 0 & \text{otherwise} \end{cases}$ Find marginal PDF and co-variance of X and Y.	3	CO2
(b)	Discuss the joint-probability distribution function (PDF) and joint-commutative distribution function (CDF), Also discuss the properties of CDF and PDF.	2	CO2
3	Attempt all parts		
(a)	Suppose that earthquakes occur with $\lambda = 2$ and with 1 week as the unit of time. (i) Find the probability that at least 3 earthquakes occur during the next 2 weeks. (ii) Find the probability distribution of the time, starting from now, until the next earthquake.	2	CO1
(b)	Let X have distribution	3	CO1

No. of pages : 03
Date:

Roll No.:

60

III SEMESTER B.Tech (IT, ITNS)
END-SEMESTER EXAMINATION 2022

Course Code-INECC06/ITECC06
Time: 3:00 Hrs

Course Title-Probability & Stoch. Proc.
Max Marks: 50

Note: - Attempt all questions in the **given order only**. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Questions	Mark s	CO
Q1	Attempt any 2 out of 3 parts		
a	In an experiment, A, B, C, and D are events with probabilities $P[A] = 1/4$, $P[B] = 1/8$, $P[C] = 5/8$ and $P[D] = 3/8$. Furthermore A and E are disjoint, while C and D are independent. i. Find $P[A \cap B]$, $P[A \cup B]$, $P[A \cap B^c]$, and $P[A \cup B^c]$. ii. Are A and B independent? iii. Find $P[C \cap D^c]$, and $P[C^c \cup D^c]$. iv. Are C^c and D^c independent?	05	CO1
b	Show that i. $P[A \cup B] \geq P[A]$ ii. $P[A \cup B] \geq P[B]$ iii. $P[A \cap B] \leq P[A]$ iv. $P[A \cap B] \leq P[B]$	05	
c	Let A and B be two events in S. The symmetric difference of A and B denoted as $A \Delta B$ is defined as $A \Delta B = (A \cup B) \cap (\overline{A \cap B})$. Show that i. $P(\overline{A} \cup \overline{B}) = 1 - P(A \cap B)$ ii. $P(A \cap B) = 1 - P(\overline{A}) - P(\overline{B})$ iii. $P(A \Delta B) = P(A \cup B) - P(A \cap B)$	05	
Q2	Attempt any 2 out of 3 parts		
a	Given a random variable X with mean μ_X, variance σ_X^2. The standardized random variable Y is defined as $Y = \frac{1}{\sigma_X} (X - \mu_X)$ Find the (i) mean of Y (ii) variance of Y and (iii) PDF of Y in terms of PDF of X.	05	CO2

The cumulative distribution function of a random variable U is

$$F_U(u) = \begin{cases} 0 & u < -5 \\ \frac{u+5}{8} & -5 \leq u < -3 \\ \frac{1}{4} & -3 \leq u < 3 \\ \frac{1}{4} + \frac{3(u-3)}{8} & 3 \leq u < 5 \\ 1 & u \geq 5 \end{cases}$$

- What is $E[U]$?
- What is $Var[U]$?
- What is $E[2^U]$?

The Gaussian random variable $X \sim \mathcal{N}(0,1)$ is transformed as follows:

$$Y = T(X) = \begin{cases} 6 & 4 \leq x \leq \infty \\ 2 & 0 \leq x < 4 \\ -2 & -4 \leq x < 0 \\ -6 & -\infty \leq x < -4 \end{cases}$$

Determine the

- PMF of Y and draw its graph
- value of $P[Y < 2]$ and
- $E[Y]$

Q3

Attempt any 2 out of 3 parts

The joint CDF of a bi-variate random variable (X, Y) is given by

$$F_{XY}(x, y) = \begin{cases} (1 - e^{-ax})(1 - e^{-by}), & x \geq 0; y \geq 0; a, b > 0 \\ 0, & \text{Otherwise} \end{cases}$$

- Find the marginal CDF's of X and Y .
- Are X and Y independent.
- Find $P(X \leq 1, Y \leq 1)$, $P(X \leq 1)$, $P(Y > 1)$ and $P(X > x, Y > y)$.

The function $h(x)$ is a monotone increasing function i.e. $h(a) \geq h(b)$, if $a > b$. Given that $Y = h(X)$. Compute the joint CDF of X, Y i.e. $F_{X,Y}(\alpha, \beta)$.

Let $Y = e^X$. Find

- The CDF of Y in terms of the CDF of X .
- The PDF, mean and variance of Y , when X is a standard Gaussian random variable (mean=0, variance=1)

Q4 Attempt any 2 out of 3 parts

a	Define a random process $Y(t) = \begin{cases} e^{-\Lambda t} \cos(\Omega t + \Theta) & , \text{ for } t \geq 0 \\ 0 & , \text{ elsewhere} \end{cases}$ where Λ , Ω and Θ are independent and uniformly distributed on $[0,1]$, $[0,2]$ and $[0,2\pi]$ respectively. Find the following i. $E[Y(t)]$ ii. $R_{YY}(t_1, t_2)$ for all t_1 and t_2 iii. Is the process wide sense stationary? Support your answer	05	CO4
b	Write a detailed note on the Gaussian Random Process. Show that if a Gaussian Random Process is WSS, then it is also a SSS process?	05	
c	Let $X(t)$ be a Poisson Process with rate λ . Find $P(X(t-d) = k X(t) = j), d > 0$.	05	

Q5 Attempt any 2 out of 3 parts

a	Explain the following Queueing models (a) M/M/1 Queueing system (b) M/M/1/N Queueing system	05	CO8
b	Define and explain Markov process. Consider the moving average of a Bernoulli sequence: $Y_n = \frac{1}{3}(X_n + X_{n-1} + X_{n-2}),$ where the X_i are independent Bernoulli sequences with $p=1/2$. Is Y_n a Markov process?	05	
c	Consider the Discrete Time Markov Chain with 3 states $\{1,2,3\}$ with state transition matrix; $P = \begin{bmatrix} 1/2 & 1/4 & 1/4 \\ 1/3 & 0 & 2/3 \\ 1/2 & 1/2 & 0 \end{bmatrix}$ i. Compute $\text{Prob}(X_3 = 1, X_2 = 2, X_1 = 3)$, if $\text{Prob}(X_1 = 1) = \text{Prob}(X_1 = 2) = 1/4$. ii. Draw the state transition diagram and comment on the states.	05	

Degree: B. Tech. Semester: 3 - Course work
END-SEMESTER EXAMINATION, Nov-Dec 2023
 Course Title: Probability Theory and Stochastic Process
 Course Code: ECECC06/EIECC06

Max. Marks: 50

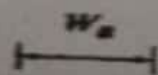
Time: 03 Hours

Note: - Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	(i) Two baskets, basket 1 and basket 2, contain, respectively, two white and one black ball, and one white and five black balls. One ball is transferred from basket 1 to basket 2 and then one ball is drawn from the latter. Suppose the drawn ball happens to be white. What is the probability that the transferred ball was black? (ii) Let X be the random variable with the following pdf. $f_X(X) = \frac{1}{2} e^{- x } \quad \text{for all } x$ Find CDF of Y if $Y=X^2$	2+3	1
1b	(i) Let X be a random variable whose probability density function is given by $f_X(x) = \begin{cases} e^{-2x} + \frac{1}{2} e^{-x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$ Write down the moment generating function for X and use the same to compute the first and second moments of X.	3+2	1
1c	(i) Suppose X has the Moment Generating Function: $m_X(t) = (1 - 2t)^{-\frac{1}{2}} \quad \text{for } t < \frac{1}{2}$ Find the first and second moments of X. (ii) Define the law of total probability.	4+1	1
Q 2	Attempt any 2 parts of the following		
2a	Let (X, Y) be a bivariate r.v. with the joint pdf $f_{XY}(x, y) = \frac{x^2 + y^2}{4\pi} e^{-(x^2 + y^2)/2} \quad -\infty < x < \infty, -\infty < y < \infty.$	2+3	2

	(i) Let X be a random variable whose probability density function is given by $f_X(x) = \begin{cases} e^{-2x} + \frac{1}{2}e^{-x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$ Write down the moment generating function for X and use the same to compute the first and second moments of X.	3+2
1c	(i) Suppose X has the Moment Generating Function: $m_X(t) = (1 - 2t)^{-\frac{1}{2}} \text{ for } t < \frac{1}{2}$ Find the first and second moments of X. (ii) Define the law of total probability.	4+1
Q 2	Attempt any 2 parts of the following	
2a	Let (X, Y) be a bivariate r.v. with the joint pdf $f_{XY}(x, y) = \frac{x^2 + y^2}{4\pi} e^{-(x^2 + y^2)/2} \quad -\infty < x < \infty, \quad -\infty < y < \infty,$ Are X and Y independent? Are X and Y uncorrelated? Justify your answer.	2+3
2b	(i) Let $X \sim \exp(\beta_1)$ and $Y \sim \exp(\beta_1)$ be independent RVs. (a) Find the joint density of X and Y (b) Calculate $P(X \leq Y)$ (c) If $Z = X - 2Y$, find $E(Z)$ and $Var(Z)$ (ii) The joint p.m.f of a bivariate r.v. (X, Y) is given by: $p_{XY}(x_i, y_j) = \begin{cases} kx_i^2 y_j, & x_i = 1, 2; y_j = 1, 2, 3 \\ 0, & \text{otherwise} \end{cases}$ Find the marginal pmf's of X and Y. Are X and Y independent?	3+2
2c	Suppose that X and Y are random variables such that $Var(X) = 9$, $Var(Y) = 4$, and $\rho_{X,Y} = 1/6$. Determine $Var(X + Y)$ and $Var(X - 3Y + 4)$.	5
Q 3	Attempt any 2 parts of the following	
3a	State and prove central limit theorem	
3b	Find the pdf $f_Y(y)$ for the following cases: (i) $Y = e^X$, and X is a uniform random variable over $(0, 1)$ (ii) $Y = X^2$, where X is a uniform random variable over $(-1, 2)$. Sketch the	1+4 2+3

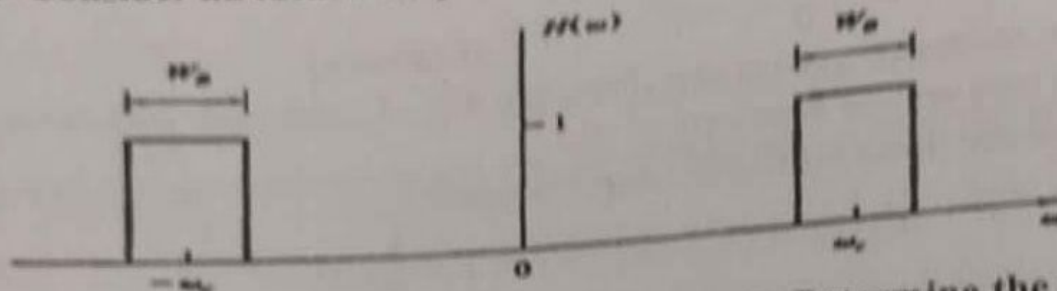
3c	Consider a random variable X having probability density function $f_X(x) = 0$ for $x < 0$. For such random variable prove, the following inequality $P(X \geq k) = \frac{E[X]}{k} \quad \text{where } k \text{ is a positive constant}$	5
Q 4	Attempt any 2 parts of the following	
4a	Define Wide Sense Stationary Process. Consider $X(t) = A \cos(\omega t + \theta)$. Is $X(t)$ a WSS process for the following cases? Justify your answer. (i) A and ω are constants and θ is a uniform random variable over $(-\pi, \pi)$. (ii) ω and θ are constants and A is a random variable.	1+2+2
4b	(i) Define Gaussian Random Process and mention at least 4 properties. (ii) Write down at least four properties of Auto correlation and cross-correlation function	5
4c	(i) Define Ergodic Random Process. Given $X(t) = A \cos(\omega t + \theta)$. A and ω are constants and θ is a uniform random variable over $(-\pi, \pi)$. Is $X(t)$ ergodic in both mean and autocorrelation? Justify your answer. (ii) Let $X(t) = A \cos(\omega t) + B \sin(\omega t)$ and $Y(t) = B \cos(\omega t) - A \sin(\omega t)$. A and B are independent random variables both with zero mean and variance σ^2 and ω is a constant. Find $R_{XY}(\tau)$.	3
Q 5	Attempt any 2 parts of the following	2
5a	(i) Distinguish between White Noise and Band-Limited White Noise with respect to their Power Spectral Density and Autocorrelation Function. (ii) Consider an ideal bandpass filter with frequency characteristics as below:	3



$H(\omega)$

Q 5
5a

σ^2 and ω is a constant. Find $R_{XY}(\tau)$.
Attempt any 2 parts of the following
(i) Distinguish between White Noise and Band-Limited White Noise with respect to their Power Spectral Density and Autocorrelation Function.
(ii) Consider an ideal bandpass filter with frequency characteristics as below:



The input to this filter is a white noise process. Determine the total noise power at the output.

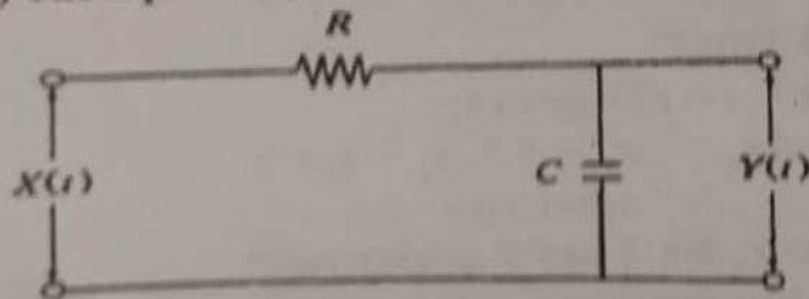
5b

Proof that if a WSS random process passing through an LTI system the output random process is also a WSS random process using suitable mathematical expressions.

5c

(i) A WSS random process $X(t)$ with autocorrelation function $R_X(\tau) = e^{-a|\tau|}$, where a is a real positive constant, is applied to the input of an LTI system with impulse response $h(t) = e^{-bt}u(t)$, where b is a real positive constant. Find $R_Y(\tau)$ for output $Y(t)$.

(ii) The input $X(t)$ to the RC filter shown below is a white noise process.



Determine the power spectrum and auto-correlation of $Y(t)$

Date:

Roll No.

THIRD SEMESTER-B.TECH.(ECE & EIOT)

MID-SEMESTER EXAMINATION, September 2022

Course Code- ECECC06/EIECC06

Course Title- Probability Theory and Stochastic Process

Time: 1:30 Hrs

Max Marks: 25

Attempt all questions. Missing data / information (if any) may be suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO
Q1	(a) A continuous random variable X has the PDF $f_X(x) = \begin{cases} \frac{1}{3}e^{-x/3}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$ Find moment generating function of X.	3	CO3
	b) Let X is a random variable and its PDF is given by: $f_X(x) = \begin{cases} \frac{1}{3} & 0 < x < 1 \\ \frac{2x^2}{7} & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$ Calculate: (i) CDF (ii) $E[X]$	2	CO1
Q2	a) A random variable X is Gaussian distributed with mean = 0 and $\sigma_X = 1$. Find $P[X > 2]$. Given $\text{erf}(1.41)=0.9533$, and $\text{erf}(1.35)=0.94376$	3	CO2
	b) The lifetime of a system expressed in weeks is a Rayleigh random variable X defined as $f_X(x) = \begin{cases} (x/200)e^{-x^2/400} & 0 \leq x \\ 0 & x < 0 \end{cases}$ What is the probability that the system lifetime will exceed one year?	2	CO1
Q3	a) Suppose the depth of water, measured in meters, behind the dam is defined by random variable having density function $f_X(x) = \left(\frac{1}{13.5}\right)u(x)e^{-(x/13.5)}$ There is an emergency overflow at the top of the dam that prevents the depth from exceeding 40.6 m. There is a pipe placed 32.0m below the overflow that feeds water to the hydroelectric generator. What is the probability that the water is wasted through emergency overflow?	3	CO2
	b) Consider the function given	2	CO2

	$F(x) = \begin{cases} 0 & \dots\dots\dots x < 0 \\ x + \frac{1}{2} & \dots\dots\dots 0 \leq x < 1/2 \\ 1 & \dots\dots\dots x \geq 1/2 \end{cases}$ Find (i) $P(X \leq 1/4)$ (ii) $P(0 < X \leq 1/4)$		
Q 4	a) Let X be a random variable with cdf $F_X(x) = \begin{cases} 1, & x \geq 1 \\ \frac{1}{2} + \frac{x}{2}, & 0 \leq x < 1 \\ 0, & x < 0 \end{cases}$ Find $P(X=0 X \leq 0.5)$	2	CO2
	b) The number of telephone calls arriving at a switchboard during any 10 min period is known to be Poisson random variable X with parameter $\lambda = 2$. Find the probability (i) that more than three calls will arrive during any 10 minute period (ii) that no calls will arrive during any 10 minute period	3	CO1
Q 5	a) Let X and Y be Joint continuous random variables with joint PDF is given below $f_{XY}(x, y) = \begin{cases} Cx + 1 & x, y \geq 0 \text{ and } x + y < 1 \\ 0 & \text{otherwise} \end{cases}$ (i) Find the constant C (ii) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$	3.0	CO2
	b) A company producing electric relays has three manufacturing plants producing 50, 30 and 20 percent, respectively, of its product. Suppose that the probabilities that a relay manufactured by these plants is defective are 0.02, 0.05 and 0.01 respectively. i. If a relay is selected at random from the output of the company, what is the probability that it is defective? ii. If a relay selected at random is found to be defective, what is the probability that it was manufactured by plant 2?	2.0	CO1

Date:

Roll No.

III SEMESTER B.Tech (IT, ITNS)
MID-SEMESTER EXAMINATION 2022

Course Code-INECC06/ITECC06

Course Title-Probability & Stochastic Processes

Time: 1:30 Hrs

Max Marks: 25

Attempt all questions. Missing data / information (If any) may be suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO
1	a) Events A, B and C are independent, with $P(A) = a$, $P(B) = b$ and $P(C) = c$. Determine the following probabilities in terms of a, b and c. i. $P(AB)$ ii. $P(A \cup B)$ iii. $P(A \cup B B)$ iv. $P(A \cup B C)$	04	CO1
	b) In the coin tossing experiment, the probability of heads (H) equals 'p' and the probability of tails (T) equals 'q'. We define a random variable X such that $X(H) = 1$ and $X(T) = 0$. Plot the CDF $F(x)$ for every x from $-\infty$ to ∞.	01	CO2
2	a) Events A and B are mutually exclusive. Determine which of the following are true and which are false. Explain the reasons. i. $P(A/B) = P(A)$ ii. $P(A \cup B C) = P(A/C) + P(B/C)$ iii. $P(A) = 0$, or $P(B) = 0$, or both. iv. $\frac{P(A B)}{P(B)} = \frac{P(B A)}{P(A)}$ v. $P(AB) = P(A)P(B)$	04	CO1
	b) State the axioms of Probability.	01	CO1

3	a) A random variable X has CDF given as $F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 - \frac{1}{4}e^{-2x} & \text{for } x \geq 0 \end{cases}$ i. Plot the CDF and identify the type of random variable ii. Find $P[X \leq 2]$, $P[X = 0]$, $P[X < 0]$, $P[2 < X < 6]$, $P[X > 10]$	04	CO2
	b) State the Poisson distribution for random variable X with parameter λ if X takes on values $0, 1, 2, \dots, \infty$. Find p_{k-1}/p_k in terms of k and λ.	01	CO2
4	a) Let X be the exponential random variable. i. Find and plot $F_X(x X > t)$. How does $F_X(x X > t)$ differ from $F_X(x)$? ii. Find and plot $f_X(x X > t)$. iii. Show that $P\{X > t + x X > t\} = P\{X > x\}$. Explain why this is called the memoryless property.	04	CO2
	b) Express $P[Y < y]$ in terms of the CDF of Y.	01	CO2
5	a) A target is made of three concentric circles of radii $3^{1/2}$, 1, and $3^{1/2}$ feet. Shots within the inner circle count 4 points, within the next ring 3 points, and within the third ring 2 points. Shots outside of the target count 0. Let R be the random variable representing distance of the hit from the centre. Suppose that the PDF of R is $f_R(r) = \begin{cases} \frac{2}{\pi(1+r^2)} & \text{for } r > 0 \\ 0 & \text{elsewhere} \end{cases}$ Compute the mean score.	04	CO2
	b) Let the mean and variance of X be m and σ^2 respectively. For what values of a and b does the random variable $Y = aX + b$ have mean ZERO and variance ONE?	01	CO2